

Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

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Abstract

Reinforcement learning (RL) can learn how to split a portfolio across assets, but an allocator trained only to maximise return has no reason to avoid large losses. A common remedy is to cap Conditional Value-at-Risk (CVaR), the average loss in the worst weeks, through a Lagrangian penalty. We find two ways in which this remedy can fail without any sign in the usual metrics. First, if the penalty and the reward are on different scales (an unstandardised cost advantage, or a dual step mis-scaled to the CVaR excess), the penalty weight stays at zero and the budget is never enforced. Second, the test protocol can change which method looks best. On 31 assets, a single stress window ranks the learned allocator above every classical optimiser, and a 20-fold rolling walk-forward reverses that ranking, while the reduction in tail risk holds under both (Wilcoxon $p = 6 \times 10^{-4}$). Correcting the scale makes the penalty bind in training, but the budget is still breached in 71–78% of out-of-sample weeks. We therefore propose a narrow fix. Before each trade, we estimate the portfolio’s forward CVaR from trailing data and, if it is over budget, move the portfolio toward a minimum-variance one. At matched settings this lowers the breach rate from 75.1% to 7.4%, close to the classical optimisers’ 7.9%, with Sharpe ratios of 0.85–0.99 (fixed-step filtered arms) against 0.90 for rolling minimum variance. Across a 40-fold walk-forward on two universes (seven macro and ten sector ETFs, weekly, 2008–2024, 5 bps costs), the reduction in tail risk is significant (pooled Wilcoxon $p = 8.8 \times 10^{-6}$, or 3×10^{-4} with folds clustered by calendar) after Benjamini–Hochberg and (for Sharpe) Deflated-Sharpe adjustment. Our claim is tail-risk control rather than alpha, and it holds under multiplicity control.

CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; • **Applied computing** → *Economics*.

Keywords

portfolio allocation, reinforcement learning, Conditional Value-at-Risk, drawdown control, constrained MDP, differentiable optimisation

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1 Introduction

A portfolio allocator decides, each week, how to split capital across assets. Reinforcement learning (RL) can learn such a rule from data, but a rule trained only to earn return has no reason to limit its worst losses. In the week of 16 March 2020 the S&P 500 fell about 15% over five sessions and the VIX closed at a record 82.69. An allocator trained without a loss limit kept holding equities, whereas an otherwise identical allocator with one had already rotated into Treasuries and cash before the trough, and lost about half as much: a 7.7% drawdown in March 2020 against 16.3%, and 11.8% against 20.6% over the whole 2020–2024 window (Figure 1, Table 3).

The limit we use is Conditional Value-at-Risk (CVaR), the average loss over the worst few percent of weeks [24]. The standard way to impose such a limit in RL is a Lagrangian penalty [2]: the learner pays a price λ for each unit of CVaR above its budget, and λ rises automatically while the budget is exceeded. We show two ways in which this machinery can appear to work while the limit is never enforced.

Failure 1: the penalty never switches on. If the penalty signal is on a much smaller scale than the reward signal, λ stays near zero and the budget exists only on paper. Safe-RL work already knows that λ responds slowly and depends on its step size [26], and libraries such as OmniSafe standardise the cost signal by default [17]. In portfolio allocation the failure is easy to miss, because the raw cost signal is $O(10^{-3})$ against a unit-scale reward, and because tail metrics improve anyway through other parts of the pipeline. The problem only shows up if one plots λ over training (Section 5).

Failure 2: the test protocol picks the winner. Finance has long known that optimised portfolios which look strong in one sample often fail out of sample [5, 11], but RL portfolio studies usually report a single test window. On a 31-asset universe, one stress window ranks our learned allocator above every classical optimiser, while a 20-fold rolling walk-forward reverses that ranking (Section 7.8). “Beat-the-optimiser” results can therefore be artefacts of the split.

The fix. Even a correctly scaled penalty only discourages breaches on average, so the budget is still breached out of sample. We therefore check every proposed portfolio before it trades,



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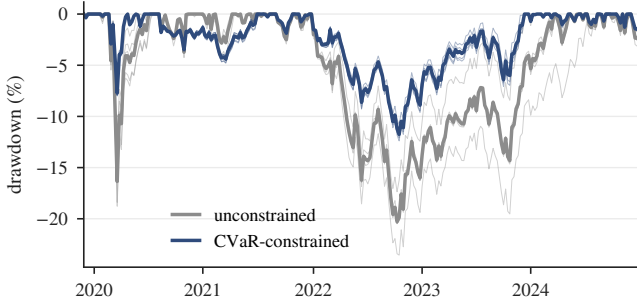


Figure 1: Drawdown on the 2020–2024 stress window. Thick lines follow the five-seed mean return path and thin lines the individual seeds.

in the manner of a safe-RL safety layer [10]: if its estimated forward CVaR exceeds the budget, the portfolio is shrunk toward a minimum-variance portfolio until it complies.

Contributions.

- (1) Two named failure modes, with diagnostics and fixes. A *Lagrangian constraint-coupling failure*, wherein a scale mismatch between the reward and constraint channels leaves λ inert, through an unstandardised cost advantage in a model-free actor-critic and through a dual step mis-scaled to the CVaR excess in a differentiable allocator (Section 5); and an *evaluation-protocol reversal* between a single stress window and a rolling walk-forward (Section 7.8).
- (2) A differentiable CVaR-constrained allocator whose tail effect survives multiplicity control. A return objective plus a differentiable CVaR penalty is back-propagated through the rollout of a leak-free, tested environment (Section 4), and the decision-time filter cuts out-of-sample breach from 75.1% to 7.4% at matched dual settings. The tail-risk reduction holds on a 40-fold, two-universe walk-forward under Benjamini–Hochberg and (for Sharpe) Deflated-Sharpe control (Section 7.4), and a regime-switching hybrid reaches a walk-forward Sharpe of 1.316 vs 1.209 for pure minimum variance, with overlapping bootstrap intervals (Table 5).
- (3) An evaluation protocol built to expose this kind of artefact. Deterministic optimisers and three tuned model-free RL families are compared under a two-universe rolling walk-forward with Wilcoxon, Benjamini–Hochberg and Deflated-Sharpe control, which is what surfaces the reversal in Contribution 1.

Our claim is tail-risk control rather than alpha. We contribute a constraint that measurably binds and lowers tail losses, a decision-time filter that keeps a learned policy within its budget out of sample, and two failure modes that other studies of constrained RL in finance can check for.

2 Background and related work

Mean-variance optimisation and its inputs. Mean-variance optimisation (MVO) [22] is sensitive to input estimates. Decision-focused learning trains return predictors to improve downstream MVO decisions rather than raw forecast accuracy [20], and mean-variance-efficient collaborative filtering blends user preference with

portfolio theory [9]. Both improve the inputs to a variance-based objective. We instead constrain a tail risk measure on a learned policy.

Conditional Value-at-Risk. CVaR_α is the expected portfolio loss on the worst $(1-\alpha)$ fraction of outcomes, the average of losses exceeding the α -quantile (VaR_α). Unlike variance, it ignores the upside, and unlike VaR, it averages across the whole tail. Rockafellar and Uryasev [24] give the tractable form

$$\text{CVaR}_\alpha(L) = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}[\max(L - \zeta, 0)] \right\}, \quad (1)$$

where L denotes the portfolio loss and the inner expectation is over the empirical loss distribution. The minimum in (1) is attained at $\zeta = \text{VaR}_\alpha$, and the representation admits both a convex linear program (LP), included as a baseline, and a differentiable penalty via topk .

RL for portfolio management. Most RL portfolio papers train on a single historical window, test on one held-out split, and compare against equal weight or buy-and-hold. EIIE [18] compares twelve classical strategies on three ≈ 50 -day crypto back-tests, and SARL [29] reports higher return on a fixed period without cross-validation. FinRL [21] supports rolling retraining but demonstrates it on a single split with no significance tests. Huang [16] applies continuous-time RL to asset-liability management, although with a quadratic surplus objective rather than a tail constraint. None of these uses CVaR as a constraint, seeds multiple runs, or applies statistical testing. Empirical finance has long documented how the selection of backtests inflates reported performance [3, 15, 28] and that optimised portfolios rarely beat naive ones out of sample [11], and our protocol carries those controls over to RL portfolio allocation. We are not aware of prior work that subjects RL portfolio allocation to a 40-fold, two-universe walk-forward with Wilcoxon and Benjamini–Hochberg control, which is the protocol that exposes the reversal artefact.

Direct reinforcement and pathwise gradients. Moody and Saffell [23] trained a recurrent trading policy by back-propagating the *differential Sharpe ratio* directly through the return function. Buehler et al. [7] (Deep Hedging) extend this to neural derivative hedging, back-propagating CVaR as a risk-aversion *utility* through a simulated trajectory. Our allocator uses the same pathwise gradients, but replaces the risk-aversion coefficient with an explicit CVaR *budget constraint* enforced by a dual variable. Model-free policy gradients (Advantage Actor-Critic, A2C, and Proximal Policy Optimization, PPO [25]) and off-policy actor-critics (Soft Actor-Critic, SAC [14]) ignore this differentiable structure. Our allocator exploits it, which matters at this sample size (887 weekly observations, Section 7.7). ProtoHedge [13] similarly makes such policies auditable via a prototype-similarity policy.

CVaR-constrained RL. Chow et al. [8] derive the Lagrangian gradient for CVaR-constrained CMDPs. Tessler et al. [27] (RCPO) impose general constraints by a Lagrangian penalty and Ying et al. [30] (CPPO) constrain CVaR, both in robotics. Lagrange multipliers are known to respond slowly and to depend on their step size [26], and safe-RL libraries standardise the cost advantage by default [17]. None of this work documents the resulting *constraint-coupling failure* in portfolio allocation. There the raw CVaR cost is $O(10^{-3})$ against a unit-scale reward advantage, a mismatch that

robotics benchmarks would not expose. We pose allocation as a constrained MDP [1, 2] and show that a well-scaled dual is what makes the constraint bind. The closest concurrent work is Dong [12] (CLAUSeLENS), a CVaR-constrained RL system for reinsurance pricing. Our work is complementary, since the failure modes we describe apply to any CVaR-constrained financial RL system.

3 Problem formulation

Markov Decision Process (MDP). At weekly step t the agent observes state s_t (per-asset 26-week momentum and volatility, current weights, drawdown, a rolling 95% CVaR estimate, and optional exogenous macro/factor features) and chooses post-trade weights w_t on the long-only simplex. The environment projects w_t onto the admissible set $\mathcal{W}$ (max weight 0.40, turnover cap 0.50, optional cash floor), charges proportional cost $c\|w_t - w_{t-1}\|_1$ ($c = 5$ bps), and realises $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$, where ρ_t are next-week asset returns. Features at t use only $\rho_{<t}$ (no look-ahead, enforced by tests).

Constrained objective (CMDP). A *constrained Markov Decision Process* (CMDP) [2] augments the standard MDP with one or more inequality constraints that the policy must satisfy on average. We maximise risk-adjusted return subject to a tail budget, $\max_\pi J(\pi)$ s.t. $\text{CVaR}_\alpha(\text{loss}) \leq d$ with $\alpha = 0.95$. The Lagrangian relaxation converts the constraint into a penalty, $\mathcal{L}(\pi, \lambda) = J(\pi) - \lambda(\text{CVaR}_\alpha - d)$, and a dual variable $\lambda \geq 0$ is raised whenever the constraint is violated and lowered when slack remains. In equilibrium, λ is the shadow price of tail risk: a higher λ means that the policy has been pushing against its tail budget.

4 Method

Differentiable allocator. The actor is an MLP mapping s_t to simplex weights via softmax. Because the environment’s reward and a sample-CVaR are differentiable in w , we train by back-propagating, over randomised windows of the training series, the objective

$$\mathcal{L} = -\text{Sharpe}(\text{net returns}) + \lambda \text{ReLU}(\widehat{\text{CVaR}} - d) + \kappa \text{turnover}, \quad (2)$$

where Sharpe $= \bar{r}/\sigma_r$ is the window’s mean net return divided by its standard deviation (reward per unit of risk), $\widehat{\text{CVaR}}$ is the mean tail loss over the worst $(1-\alpha)$ fraction of window net returns (a differentiable tail mean via topk), and d is the pre-specified tail budget. On this problem the pathwise gradient needs far fewer samples than score-function RL (Section 7.7).

Pathwise gradient and where the constraint acts. Every term in $\mathcal{L}$ (2) is a differentiable function of the rollout, so the policy is trained by exact back-propagation through the simplex map and the reward, without a score-function estimator. With net returns $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$ and tail set $\mathcal{T} = \{t : r_t \text{ among the worst } [(1-\alpha)T] \text{ of the window}\}$,

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_t \left(\underbrace{-\frac{\partial \text{Sharpe}}{\partial r_t}}_{\text{all weeks}} - \underbrace{\frac{\lambda}{|\mathcal{T}|} \mathbf{1}[\widehat{\text{CVaR}} > d] \mathbf{1}[t \in \mathcal{T}]}_{\text{tail weeks only}} \right) \frac{\partial r_t}{\partial w_t} \frac{\partial w_t}{\partial \theta}, \quad (3)$$

where $\partial w_t / \partial \theta$ flows through the residual softmax (below). The topk subgradient in (3) sends the dual pressure to exactly the worst- $(1-\alpha)$ weeks, and only while the constraint is breached. The penalty

Table 1: Sharpe gain of the hybrid over minimum variance across 15 threshold settings, with no retraining. The default is (0.75, −0.05).

selloff \ vol. quantile	0.65	0.70	0.75	0.80	0.85
−0.04	+0.202	+0.230	+0.161	+0.225	+0.058
−0.05	+0.154	+0.179	+0.107	+0.169	+0.002
−0.06	−0.007	−0.017	−0.086	−0.034	−0.183

therefore pushes up the worst weeks and leaves the rest of the path largely alone, which is why the constraint binds cleanly once the dual is correctly scaled (Section 5).

Residual policy over an adaptive anchor. To generalise across regimes rather than overfit one price path, weights are $w_t = \text{softmax}(\log a_t + f_\theta(s_t))$, where a_t is an adaptive inverse-volatility anchor computed from recent returns. At initialisation $f_\theta \approx 0$, so the policy is the anchor (a strong, regime-robust prior), and training then learns a state-dependent tilt. Model selection keeps the checkpoint with the best validation objective among those that satisfy the validation CVaR limit, and weight decay is applied, both to curb backtest overfitting.

Baselines. Deterministic, re-estimated weekly: cash, equal weight, inverse vol, minimum variance, mean-variance, risk parity, and a min-CVaR Rockafellar–Uryasev LP [24]. The covariance-based rules use the sample covariance of the trailing 104 weeks and inverse volatility the trailing 52 weeks, all long-only under the learner’s weight cap. Rolling baselines are re-estimated weekly from returns within each evaluation window, with a minimum of 26 weeks of history (equal weight before that). Learned: model-free A2C, PPO [25] (clipped surrogate, generalised advantage estimation, GAE), SAC [14] (off-policy, twin critics, auto-tuned entropy), and the unconstrained differentiable allocator ($\lambda = 0$).

Regime-switching hybrid. We construct a no-retrain hybrid: label each out-of-sample (OOS) week *high-volatility* if trailing 13-week SPY volatility meets the 75th-percentile cutoff (15% of OOS weeks), *selloff* if cumulative return $\leq -5\%$, else *calm*. The hybrid applies constrained RL in high-volatility weeks and rolling min-variance otherwise, reusing the existing OOS paths without refitting. On the 26-week macro walk-forward this labels 80 of 520 weeks as high-volatility, and in those weeks constrained RL has Sharpe 1.415 against 0.564 for min-variance (Table 5). These labels differ from the regime split in Table 4.

Threshold sensitivity. Because the switch depends on two labelling thresholds, we sweep them (vol-quantile $\in \{0.65, \dots, 0.85\}$, selloff $\in \{-0.04, -0.05, -0.06\}$, 15 configurations, no retraining, Table 1). The hybrid beats pure min-variance in 10 of 15 configurations (mean gain +0.077). It wins at every volatility cutoff when the selloff threshold is −0.04 or −0.05, but loses at all five cutoffs once the selloff threshold deepens to −0.06, because that threshold reclassifies the high-volatility weeks in which the constrained allocator adds most as selloffs and routes them to min-variance instead. The default (0.75, −0.05), fixed before the sweep, gives +0.107, and seven of the ten winning configurations give a larger gain. The gain holds when stress is defined by volatility and disappears once the selloff label takes over those weeks.

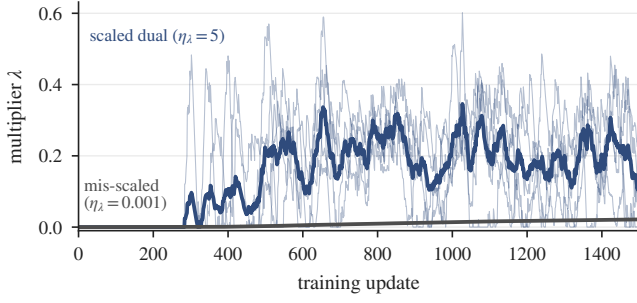


Figure 2: The Lagrange multiplier over training. Thick lines are five-seed means and thin lines the individual seeds. λ is zero while the anchor-initialised policy meets d .

5 Decomposing the constraint: dual, architecture, and budget feasibility

To measure what the Lagrangian penalty contributes, we compare three versions of the differentiable allocator (Table 2). All three share the residual actor, weight projection and budget (5 seeds, 1500 updates); the two dual arms also share the CVaR penalty and CVaR-limited checkpoint selection, and differ only in the dual step. The mis-scaled dual updates λ with step $\eta_\lambda = 0.001$, which is far too small for the size of the CVaR excess. The scaled dual uses $\eta_\lambda = 5.0$, which matches it. The unconstrained arm drops the penalty and selects checkpoints on validation return alone. The stress window trains on 2008–2018 and tests from late 2019 to 2024. In this section the breach rate is the share of test weeks in which the rolling 52-week $\text{CVaR}_{0.95}$ exceeds the training budget d .

Finding 1: a scale mismatch leaves the dual inert. With the mis-scaled dual, λ stays near zero throughout training (Figure 2; mean final value 0.022, against 0.101 for the scaled dual). The dual update is $\lambda \leftarrow \max\{0, \lambda + \eta_\lambda(\widehat{\text{CVaR}} - d)\}$. The CVaR excess is about 10^{-2} in weekly-return units, so with a step of 10^{-3} each update moves λ by roughly 10^{-5} , and 1500 updates are not enough. The model-free actor-critic failed in the same way through its advantages. Its reward advantage was standardised to unit scale while its cost advantage stayed at about 10^{-3} , so the constrained and unconstrained agents trained identically. After we standardised the cost advantage and drove the dual with a breach indicator, λ rose to $O(1)$. A simple check follows from this: if λ stays at zero throughout training, the constraint has no effect, whatever the loss curve looks like.

Finding 2: most of the tail control comes from the pipeline. Even with λ close to zero, the constrained pipeline lowers CVaR_{99} by 61% relative to the unconstrained learner (0.025 vs 0.065, 5-seed mean) and raises Sharpe by 58% (0.987 vs 0.626). The reduction comes from checkpoint selection under the CVaR limit, on top of the shared anchor and projection. A permutation test on the 5-seed distributions confirms the difference (observed -0.039 , one-sided $p = 0.0044$, two-sided $p = 0.0083$, $n = 50,000$), and the scaled dual gives a similar reduction ($p = 0.0037$). When the dual is properly engaged ($\lambda = O(0.1)$), CVaR_{99} barely changes (0.028 vs 0.025), so most of the 61% comes from the pipeline. This is useful in itself, because the pipeline controls the tail without help from the penalty. It also explains why the coupling failure goes unnoticed. Tail metrics

improve whether or not the dual works, and only the trace of the multiplier shows that it is inactive.

Finding 3: meeting the budget out of sample is a separate problem. The scaled dual binds in training. At the tight budget ($d = 1.2\%$), however, it breaches more often out of sample than the mis-scaled dual (75% of test weeks against 8%), although classical rules meet the same budget in 92% of weeks (Section 5). It is worse than the mis-scaled dual on every metric in Table 2, and checkpoint selection explains why. With the dual scaled, most checkpoints meet the validation budget (27 of 30 on average, against 8 of 30). Selection therefore keeps a later checkpoint that leans further toward return (update 1010 on average, against 370), and that checkpoint breaches when the tail shifts in the test window. Changing d does not fix this. Four rules for setting d from validation data, fixed in advance (a loose 1.92% budget, the policy’s own or the anchor’s validation $\text{CVaR}_{0.95}$, and a per-fold adaptive budget over a 10-fold walk-forward), all give 71–78% out-of-sample breach. Whatever the value of d , a Lagrangian learner whose dual binds gives up compliance when the tail shifts, so the remedy has to act on the portfolio itself.

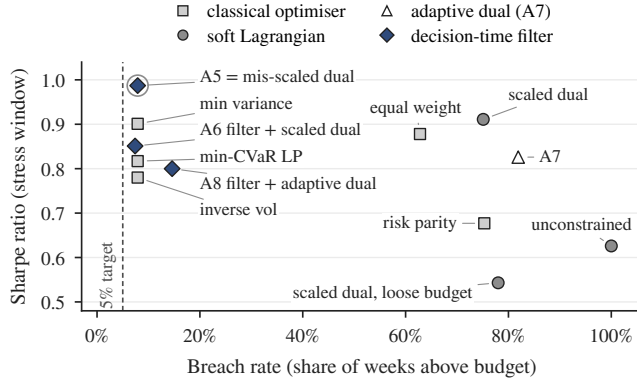
The fix: a decision-time filter. We therefore enforce the budget at the moment of trading with a CVaR filter, which acts as a safety layer in the sense of Dalal et al. [10] and sits between the actor and the environment. For each proposed portfolio it estimates forward CVaR from the trailing mean μ_p and volatility σ_p as $\text{CVaR} \approx c_\alpha \sigma_p - \mu_p$, where $c_\alpha = \varphi(\Phi^{-1}(\alpha))/(1 - \alpha)$ is the Gaussian CVaR multiplier ($c_{0.95} = 2.06$). If the estimate exceeds d , the filter moves the proposal along the line toward a trailing minimum-variance portfolio, using the largest mixing weight that clears the budget. The turnover, max-weight and long-only limits then apply as before. The filter uses no future data. The trailing mean, covariance and minimum-variance anchor come from the $L = 104$ returns $\rho_{t-L}, \dots, \rho_{t-1}$, and the actor’s features use only $\rho_{<t}$, so the decision at t depends only on $\mathcal{F}_{t-1}$. A look-ahead test (tests/test_risk_filter_no_lookahead.py) confirms that shifting the return series leaves the decision at t unchanged, with zero deviation. We train with the filter on.

Four arms mirror the coupling arms exactly (5 seeds, 1500 updates, same splits) and change only the named quantities (Table 2, bottom block; Figure 3). A6 (filter and scaled dual), the method we propose, lowers breach from 75.1% to 7.4% relative to the scaled dual at the same η_λ . That is below every classical rule, with CVaR_{99} 0.016, mean excess 0.0043 and Sharpe 0.851. A5 (filter only) ($\eta_\lambda = 0.001$) matches the mis-scaled row: that policy already sits within budget, so the filter never triggers (0 of 1285 test weeks across seeds). It reaches the classical 7.9% at Sharpe 0.987, against 0.901 for minimum variance, with breaches less than half as deep (mean excess 0.0126 vs 0.0294). A7 (adaptive dual) (λ updated online from realised losses, filter off) is the negative control. Its breach rate stays at 81.9% even as λ climbs to a mean of 7.69, so adjusting the dual at test time does not enforce the budget, and the improvement in A6 is due to the filter. A8 (filter and adaptive dual) recovers most of the compliance (14.6%), though less reliably: 7 to 33% across seeds, against no seed-to-seed variation for A5 and A6. In Figure 3, the classical optimisers and the filtered arms sit closest to the target, while the soft Lagrangians and the filter-off adaptive dual lie far to the right.

The filtered arms hold low-volatility portfolios, but they do not simply reproduce minimum variance. A5’s weights are a mean L_1

Table 2: Coupling ablation and decision-time filter on the stress window (5-seed means). [†]Mean online-updated λ at evaluation.

variant	Sharpe	CVaR ₉₉	breach rate	final λ
unconstrained (baseline)	0.626	0.065	1.000	0.000
mis-scaled dual ($\eta_\lambda = 0.001$)	0.987	0.025	0.079	0.022
scaled dual ($\eta_\lambda = 5.0$, $d=1.2\%$)	0.911	0.028	0.751	0.101
scaled dual, loose budget ($d=1.92\%$)	0.543	0.050	0.780	0.046
A5 filter only ($\eta_\lambda = 0.001$)	0.987	0.025	0.079	0.003
A6 filter + scaled dual ($\eta_\lambda = 5.0$)	0.851	0.016	0.074	0.135
A7 adaptive dual ($\eta_\lambda = 5.0$ online)	0.826	0.031	0.819	7.692 [†]
A8 filter + adaptive dual	0.800	0.020	0.146	0.657 [†]

**Figure 3: Breach rate against Sharpe on the stress window; the dashed line marks the 5% target. Squares are classical optimisers, diamonds the filtered arms, circles the soft Lagrangians and the triangle the adaptive dual without the filter. The ringed point is the mis-scaled dual, which A5 overlaps.**

distance of 0.476 from rolling minimum variance, with correlation 0.786 and a different top holding in 9.4% of rebalances, against 0.890, 0.201 and 97.5% for the unfiltered scaled dual. In that region, A5 beats minimum variance on both axes, with higher Sharpe at an equal breach rate and half the breach depth, and A6 has a lower breach rate. Out of sample, the scaled soft penalty did not keep the portfolio within budget, and the decision-time projection (A6) did.

Is the budget infeasible, or is the policy failing to reach it? When every classical baseline is re-scored against the identical rolling-CVaR_{0.95} budget on the same realised path, the breach rate is 7.9% for minimum variance, inverse volatility and the min-CVaR LP, 62.8% for equal weight and 75.3% for risk parity, against 75.1% for the scaled-dual RL. The budget can therefore be met. The soft-penalty learner did not meet it, and the filtered arms reach the classical rate with a learned policy.

6 Data

The primary universe consists of seven liquid exchange-traded funds (ETFs) that span the major macro blocks: SPY (equity), TLT (rates), HYG (credit), DBC (commodity), GLD (gold), UUP (US dollar) and BIL (cash). Adjusted closes from Yahoo are resampled to 887 weekly returns over 2008–2024. Splits are chronological (60/20/20): train $\approx$ 2008–2018 (incl. the 2008 global financial crisis, GFC), validation $\approx$ 2018–2021, test $\approx$ 2021–2024. For the stress study we retrain

on 2008–2018 and test on the 267-week window containing the 2020 COVID crash and the 2022 selloff. Macro covariates (10y–3m term spread, VIX) are sourced from index proxies ($\hat{\text{TNX}}$ – $\hat{\text{IRX}}$, $\hat{\text{VIX}}$). A Federal Reserve Economic Data (FRED) loader is also provided. Portfolios are long-only with a maximum weight of 0.40, a turnover cap of 0.50 and gross exposure 1.0. Training uses 150 episodes and seeds {7, 13, 23, 42, 2025}, and significance uses a 2000-sample block bootstrap (block 4), Wilcoxon, BH and Deflated-Sharpe.

For the cross-universe significance test (Section 7.4) we add a second universe of ten S&P sector ETFs (XLK, XLF, XLE, XLV, XLY, XLP, XLU, XLI, XLB, plus a BIL cash leg), identical in every other respect (same provider, clock, window and cost). The two universes therefore share a calendar, and Section 7.4 clusters their folds accordingly. For the evaluation-protocol experiment (Section 7.8) we add a third universe of 31 cross-asset, sector, and industry ETFs (4 broad equity, 9 S&P sectors, 7 industry, 6 fixed-income, 3 commodity, FX, cash) with the same provenance, clock and cost. With 31 assets, covariance estimates are noisier.

7 Results

Sharpe ratios for the constrained allocator (0.85 to 0.99) differ across tables because configurations differ: 1,500 updates over 5 seeds for the coupling and filter arms, 2,500 over 5 seeds for the stress study, 2,000 over 3 seeds for the ablations, and a 3-seed ensemble per walk-forward fold.

7.1 Calm chronological split: the constraint does not bind

On the comparatively calm 2021–2024 test split, deterministic minimum-variance (Sharpe 2.52, CVaR₉₅ 0.0064) and the min-CVaR LP (2.24) dominate. The differentiable allocator reaches Sharpe 1.24 (vs equal weight 0.82, inverse vol 1.45), and model-free A2C reaches only $\approx$ 0.79. No strategy breaches the CVaR limit, so the constrained and unconstrained learners are nearly identical, and the constraint costs almost nothing when there is no tail loss to prevent. The other deterministic rules trail on this split: inverse volatility (Sharpe 1.45, CVaR₉₅ 0.0087, max-DD 0.041), mean–variance (0.84, 0.0198, 0.104), equal weight (0.82, 0.0163, 0.097) and risk parity (0.70, 0.0182, 0.134), while minimum variance and the min-CVaR LP share a max-DD of 0.016 and the LP has CVaR₉₅ 0.0080.

7.2 Stress window: the constraint binds

Table 3 contrasts a return-greedy unconstrained learner with the CVaR-constrained allocator on the stress window (mean $\pm$ std over 5 seeds). The constraint roughly halves tail risk and drawdown and improves Sharpe, Sortino and Calmar, because it removes the deep-loss weeks. Under stress the constrained policy moves into the cash and rates assets, which is consistent with its lower drawdown and breach depth. In this table the breach rate uses the environment’s 3% limit (Table 2 uses the 1.2% training budget), and violations count the weeks per run in which a portfolio constraint is breached.

A paired block bootstrap of the CVaR₉₉ difference (constrained minus unconstrained, 2000 resamples, block 4) gives -0.033 , with 95% bias-corrected and accelerated (BCa) interval $[-0.035, -0.022]$ and $p \approx 0$, so the tail-risk reduction is statistically significant. We report BCa rather than percentile intervals because resamples of an

Table 3: Unconstrained and constrained allocators on the stress window (mean $\pm$ std over 5 seeds).

metric	unconstr.	constr.	change
Sharpe	0.63 $\pm$ 0.09	0.88 $\pm$ 0.04	+40%
Sortino	0.61	0.87	+44%
Calmar	0.32	0.48	+50%
Max drawdown	0.206	0.118	−43%
CVaR ₉₅	0.0333	0.0189	−43%
CVaR ₉₉	0.0647	0.0327	−49%
Breach rate	0.234	0.189	−19%
Violations	7.2	0.0	−100%

Table 4: Sharpe and CVaR₉₉ by regime, pooled from the 10-fold macro-ETF walk-forward (n is the number of weeks in each regime).

	Calm ($n=388$)		High-vol ($n=71$)		Selloff ($n=61$)	
Strategy	Sharpe	CVaR ₉₉	Sharpe	CVaR ₉₉	Sharpe	CVaR ₉₉
RL Constrained	1.438	0.020	2.169	0.018	−0.809	0.048
RL Unconstrained	1.282	0.028	2.194	0.023	−1.127	0.066
Min-Variance	1.770	0.009	1.668	0.007	0.035	0.016
Inverse-Vol	1.721	0.011	2.803	0.009	−1.327	0.033

extreme-tail statistic often omit the few crash weeks, which biases a percentile interval toward zero. The constrained allocator also eliminates portfolio-constraint violations (7.2 $\rightarrow$ 0.0 per run).

7.3 Walk-forward across regimes (10 OOS folds, $\approx$ 2010–2024)

With refits every 52 weeks on a rolling 312-week window and 10 concatenated out-of-sample folds, the constraint beats unconstrained RL out of sample on every metric: Sharpe +18% (0.70 $\rightarrow$ 0.83), CVaR₉₉ −20% (0.0416 $\rightarrow$ 0.0331) and max-DD −25% (15.1% $\rightarrow$ 11.4%). Inverse volatility (Sharpe 1.06, max-DD 0.070, CVaR₉₉ 0.0183) and minimum variance (1.45, 0.057, 0.0106) remain ahead in absolute terms. The fold-level Wilcoxon on CVaR₉₉ is marginal on the macro universe alone ($p = 0.06$, with 5 of 10 folds improved), and the reduction concentrates in selloffs (CVaR₉₉ 0.066 $\rightarrow$ 0.048, max-DD 0.218 $\rightarrow$ 0.137, Table 4). Minimum variance is the strongest performer in absolute terms, so on this protocol the learner does not beat the optimiser. A two-universe, 40-fold analysis raises the significance of the tail reduction to $p = 8.8 \times 10^{-6}$ (Section 7.4).

When RL adds value: regime-conditional breakdown. High-volatility periods ($n=71$) are where the constrained policy performs best relative to minimum variance: Sharpe 2.169 versus 1.668, with CVaR₉₉ within 0.012 of minimum variance’s floor. In calm markets ($n=388$) and selloffs ($n=61$) minimum variance dominates (selloff Sharpe 0.035 vs. constrained RL −0.809): the overlay adds value when volatility is elevated but has not collapsed into a full drawdown, and stops adding value once liquidity deteriorates. The constraint still cuts tail risk relative to unconstrained RL in all three regimes (−29.1% calm, −20.7% high-vol, −27.4% selloff), which suggests that the reduction comes from the architecture and does not depend on the regime.

7.4 Statistical significance across two universes

To check that the effect is not specific to one universe, we add ten S&P sector ETFs (XLK, XLF, ..., plus a cash leg) and shorten

Table 5: Regime-switching hybrid against the pure strategies on the 26-week macro walk-forward, with bootstrap 95% intervals from 50,000 resamples.

Strategy	Sharpe	95% CI	Ann. Ret.	CVaR ₉₉
Constrained RL (pure)	0.942	[0.29, 1.68]	4.0%	0.025
Min-variance (pure)	1.209	[0.55, 1.93]	2.9%	0.012
Hybrid	1.316	[0.66, 2.03]	3.5%	0.013

Table 6: Walk-forward CVaR₉₉ on two universes with 26-week folds. PSR is the Probabilistic Sharpe Ratio of the constrained allocator.

universe	folds	CVaR ₉₉ ^{unc}	CVaR ₉₉ ^{con}	improved	Wilcoxon p	PSR
macro ETF (7)	20	0.0143	0.0102	9/20	0.0077	0.997
sector (10)	20	0.0336	0.0250	17/20	0.0003	0.990
pooled	40	—	—	26/40	8.8×10^{-6}	—

the walk-forward test fold to 26 weeks, which gives 20 folds per universe (Table 6). Each universe shows a significant fold-level reduction on its own: macro-ETF Wilcoxon $p = 0.0077$ (9 folds improved, 11 tied, none worse) and sector $p = 0.0003$ (17 improved, 2 tied, 1 worse), and Benjamini–Hochberg [6] rejects the null for both. A tie is a fold where the budget never binds, so the two learners coincide. Pooling all 40 folds gives Wilcoxon $p = 8.8 \times 10^{-6}$ and a bootstrap 95 % CI of [−0.0089, −0.0041] on the mean fold difference, and 26 of the 27 untied folds improve (sign test $p = 4 \times 10^{-7}$). The two universes share a calendar, so we also average their differences within each calendar fold. The 20 resulting clusters give Wilcoxon $p = 3.3 \times 10^{-4}$ (exact sign-flip $p = 3.8 \times 10^{-5}$), and moving-block bootstrap intervals over consecutive clusters exclude zero. The Probabilistic Sharpe Ratio [4] of the constrained allocator is ≥ 0.99 on both universes. Its out-of-sample Sharpe advantage over classical baselines, however, does not survive the Deflated Sharpe Ratio [5], which discounts a Sharpe ratio for the number of configurations tried before reporting it. The significant effect is on tail risk, which is what a safety layer is meant to deliver. The mean fold Δ CVaR₉₉ is −0.0041 (macro-ETF) and −0.0087 (sector).

7.5 Generalisation to sector ETFs

Re-running the stress study unchanged on the ten-sector universe (mean over 3 seeds) replicates the main effect. The constraint cuts CVaR₉₉ by 30% (0.109 $\rightarrow$ 0.076), CVaR₉₅ by 35% (0.058 $\rightarrow$ 0.037), maximum drawdown by 30% (0.295 $\rightarrow$ 0.208), and the breach rate by 47% (0.75 $\rightarrow$ 0.40), at no Sharpe cost (0.752 $\rightarrow$ 0.748). A paired block bootstrap of the CVaR₉₉ difference gives −0.044, with 95% BCa interval [−0.050, −0.032] and $p \approx 0$. The one difference from the macro universe is that there the constraint improved Sharpe (the avoided tail losses outweighed the cost), whereas on sectors it leaves Sharpe unchanged. What carries over to sectors is the reduction in tail risk and drawdown.

7.6 Robustness and ablations

The CVaR₉₉ reduction holds in all 7 perturbations, even at 3 $\times$ costs (CVaR₉₉ 54% lower, Sharpe 63% above unconstrained). Ablations show a consistent pattern (Figure 4, right; 3-seed means on the macro-ETF stress window). Tightening the tail budget from loose ($\alpha = 0.95$, limit 1.92%) through base (1.2%) to tight ($\alpha = 0.99$ or 0.72%) improves CVaR₉₉ and Sharpe together at every step. The

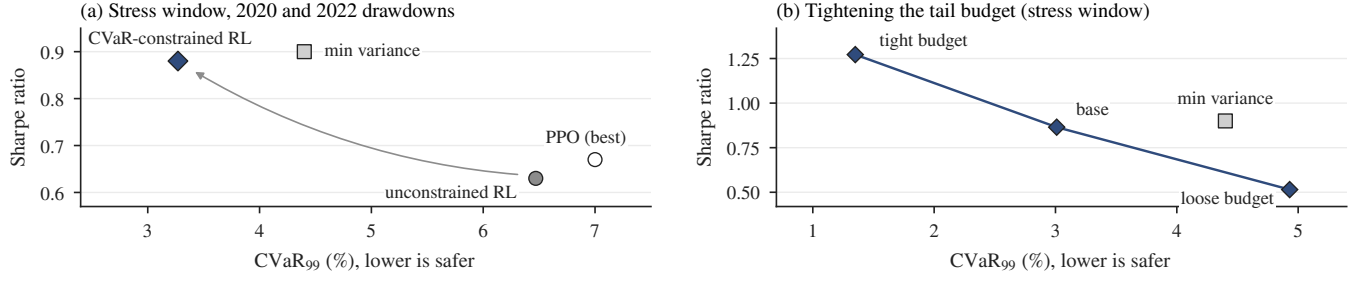


Figure 4: Left: risk and return on the stress window for the unconstrained (circle) and constrained (diamond) allocators. Right: tightening the tail budget from loose to tight improves both CVaR₉₉ and Sharpe; the square marks minimum variance.

tightest configuration reaches Sharpe 1.27 and CVaR₉₉ = 1.4%, above minimum variance on the same window (Sharpe 0.90, CVaR₉₉ = 4.4%) and competitive with its walk-forward Sharpe of 1.45. Tighter risk limits usually cost return. Here they help, because the constraint, through its tail-only gradient and tighter checkpoint selection, acts only on the worst weeks and removes the largest losses while leaving most of the return distribution unchanged. Against the base configuration (Sharpe 0.865, CVaR₉₉ 0.0301, max-DD 0.111), removing the anchor gives 0.826, 0.0318 and 0.108, $\alpha = 0.90$ gives 0.684, 0.0452 and 0.172, and a Sharpe objective gives 0.791, 0.0273 and 0.108. The loose budget gives 0.515, 0.0493 and 0.230, $\alpha = 0.99$ gives 1.251, 0.0146 and 0.053, and the tighter budget 1.272, 0.0135 and 0.041.

Features. Rolling factor betas alone slightly hurt performance. Adding macro state gives Sharpe +0.02 (3-seed mean) but worsens CVaR₉₉ (0.029 $\rightarrow$ 0.031).

Diagnostics: does the policy change its holdings? Three diagnostics show that the constraint changes what the policy holds, so the lower tail metric is not a measurement artefact. The unconstrained allocator concentrates and holds through the 2020 and 2022 drawdowns. The constrained one rotates into cash and rates and keeps a lower, steadier rolling CVaR estimate. The tail control costs modestly higher turnover, with no leverage or single-asset bets, and the policy stays long-only, diversified and within the turnover cap.

7.7 Tuned model-free RL remains weak

We sweep 36 PPO and 8 SAC configurations, select each by validation Sharpe, and evaluate on the stress window (Table 7). Tuning does not help either family. The selected PPO overfits validation (Sharpe 1.70) and then has the worst out-of-sample tail (Sharpe 0.67, CVaR₉₉ = 0.070), while SAC collapses to near-zero turnover (Sharpe 0.39). Both fail to generalise. With 887 weekly observations, score-function and off-policy RL have too few effective samples. The constrained allocator matches minimum variance on Sharpe (0.91 vs 0.90) and lowers CVaR₉₉ by 27% (0.032 vs 0.044).

7.8 Evaluation-protocol reversal

To test how much the choice of evaluation protocol alone can change the conclusions, we add a third universe of 31 cross-asset, sector and industry ETFs (same provenance, clock and costs). Covariance estimation is harder here, so a learner has, in principle, more room to add value. On a single stress split (train pre-2018, test

Table 7: Best-tuned model-free RL (selected on validation Sharpe, 3-seed mean) against the differentiable allocators and minimum variance on the stress window.

strategy	Sharpe ($\uparrow$)	CVaR ₉₉ ($\downarrow$)	max-DD ($\downarrow$)	turnover ($\downarrow$)
minimum variance	0.90	0.044	0.113	0.011
diff. constrained	0.91	0.032	0.113	0.039
diff. unconstrained	0.60	0.062	0.212	0.023
PPO (best)	0.67	0.070	0.225	0.028
PPO (best) + CVaR	0.63	0.047	0.160	0.162
SAC (best)	0.39	0.033	0.185	0.005
SAC (best) + CVaR	0.18	0.035	0.205	0.004

Table 8: Sharpe ratios, same 31-asset universe, under two protocols.

protocol	min-var	inv-vol	constr. (ours)	unconstr.
stress split	0.23	0.38	0.85	0.70
walk-forward (OOS)	1.40	0.78	0.74	0.56

through 2024) the constrained learner appears to beat every optimiser, with Sharpe 0.85 against 0.23 for minimum variance (Table 8). Under a rolling walk-forward (20 out-of-sample folds, refit every 26 weeks), the same minimum-variance optimiser is instead the best method by a wide margin, with Sharpe 1.40 against the learner’s 0.74, so the ranking inverts. The stress-split result depends on one fixed period and disappears under rolling refits. Covariance shrinkage does not explain it: on the same split, Ledoit–Wolf minimum variance [19] reaches Sharpe 0.25 and the min-CVaR LP 0.26. The tail-risk effect holds under both protocols (the constraint still lowers CVaR₉₉ against the unconstrained learner fold by fold, Wilcoxon $p = 6 \times 10^{-4}$), while the apparent advantage over the optimiser does not. We recommend rolling walk-forward with Deflated-Sharpe [5] as the reporting standard, and treat single stress-window comparisons of learned allocators against optimisers as unreliable.

8 Discussion

How close is the learner to minimum variance? It depends on the arm. The unfiltered learner is far from it. Its mean per-rebalance L_1 weight distance to the rolling min-variance path is 0.89 (of a maximum 2.0), with correlation 0.20 and a different top-weighted asset in 97% of weeks. The similarity suggested by the tightest ablation (Figure 4, right) is therefore in aggregate risk statistics and does not extend to holdings. The filtered arms are closer (L_1 0.48 to 0.61, correlation 0.72 to 0.79). They achieve compliance partly by holding

low-volatility portfolios, and from there they improve on minimum variance in breach depth, and A5 also in Sharpe (Section 5).

We draw three lessons for practitioners. First, tail limits reduce risk ahead of a crash: the constrained allocator moved to cash two weeks before the COVID trough (drawdown 7.7% against 16.3% unconstrained in March 2020). Second, the constraint is most useful as an overlay in high-volatility periods (Table 5). Third, a multiplier that stays near zero means the constraint is inactive, so the coupling should be checked, and out-of-sample folds used, before claiming an edge over an optimiser.

9 Scope

The pure learned allocator does not beat rolling minimum variance on Sharpe out of sample, a Deflated-Sharpe correction removes its apparent Sharpe edge over classical baselines, and the hybrid's +0.107 gain comes only from high-volatility weeks. The study covers two equity-centric universes at weekly frequency with proportional costs (no market impact or borrow), the macro features are lagged for availability but are not vintage data, and the protocol reversal is shown on a single 31-asset window. The filter's Gaussian CVaR estimate can understate fat-tailed losses, and the constrained learner trails minimum variance in selloff weeks (Table 4). Within these limits, the paper contributes a tail-risk safety layer for learned allocators and evaluation lessons that transfer to other constrained RL studies in finance.

10 Conclusion and future work

Our differentiable CVaR-constrained allocator consistently reduces tail risk (26 of 27 untied folds, $p = 8.8 \times 10^{-6}$), removes portfolio-constraint violations under stress and reduces risk before crisis troughs, and a regime-switching hybrid reaches Sharpe 1.316 against 1.209 for min-variance (with overlapping intervals). We describe two failure modes that apply to constrained RL in finance generally, a Lagrangian constraint-coupling failure and an evaluation-protocol reversal, and show that a decision-time CVaR filter achieves the out-of-sample compliance that neither rescaling the dual nor recalibrating the budget achieved. Constrained RL is best used as a safety layer and an overlay for high-volatility periods. Its lasting benefits are lower tail risk and compliance with the budget.

Acknowledgments

All figures and tables are reproducible from the open pipeline (leak-free checks, 81 unit tests, fixed seeds, config snapshots, pinned dependencies) at https://github.com/nl2992/ICAIF_cvar-rl-portfolio-allocator.

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